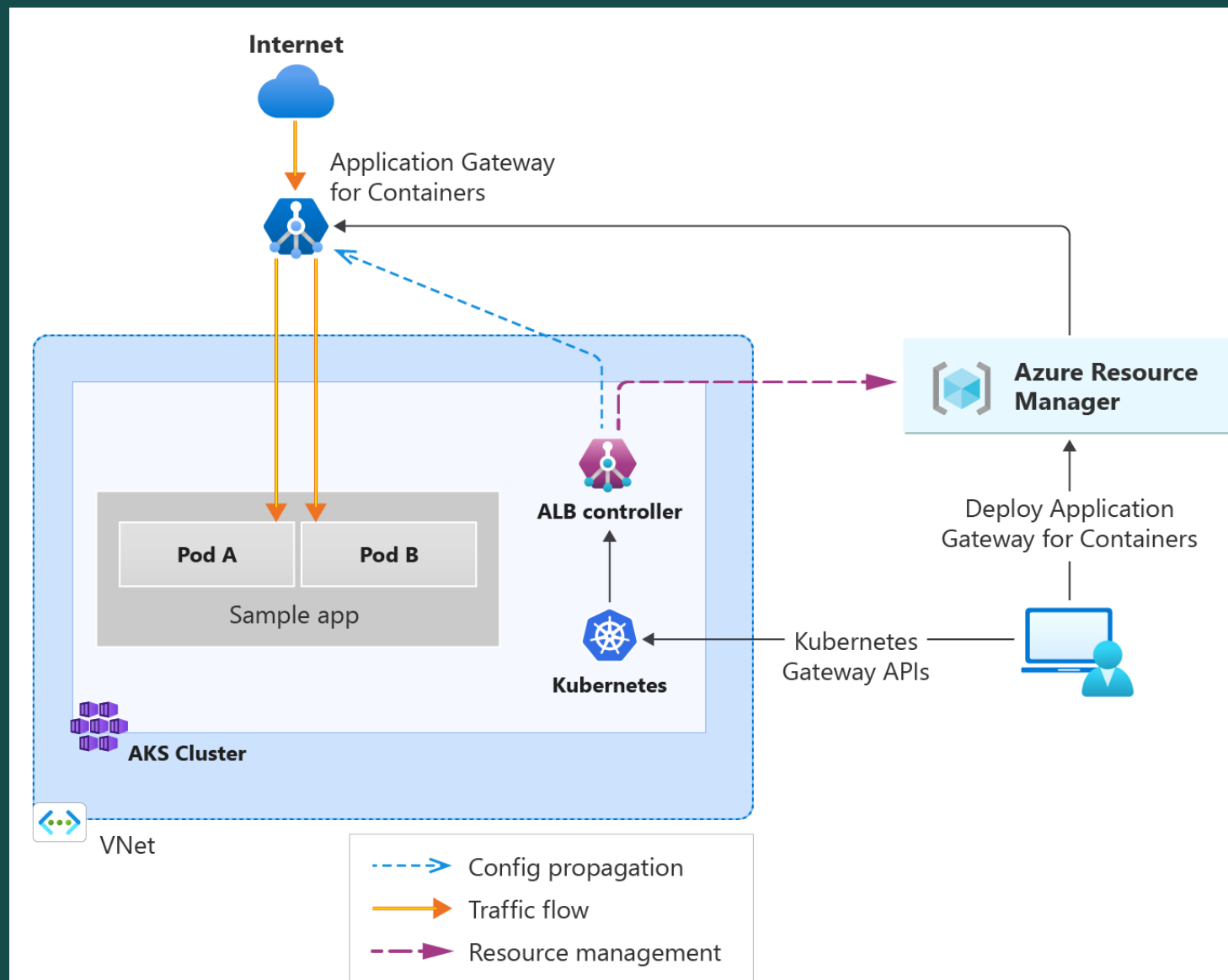




What is Application Gateway for Containers?



- **Application Gateway for Containers is a layer 7 (application) load balancing and traffic management solution for Kubernetes workloads**
- Part of the Application Gateway product family and **evolution of the Application Gateway Ingress Controller (AGIC)**
- Application Gateway for Containers is that it **supports both the Ingress and Gateway APIs**
- Because Application Gateway for Containers supports Gateway API, it can may have **additional features** like:
 - Header and query string based routing
 - Traffic splitting
 - Mutual authentication (MTLS) to backend targets
 - Flexible deployment strategies
 - Improved performance
 - More





What is Application Gateway for Containers?



Private IP addresses

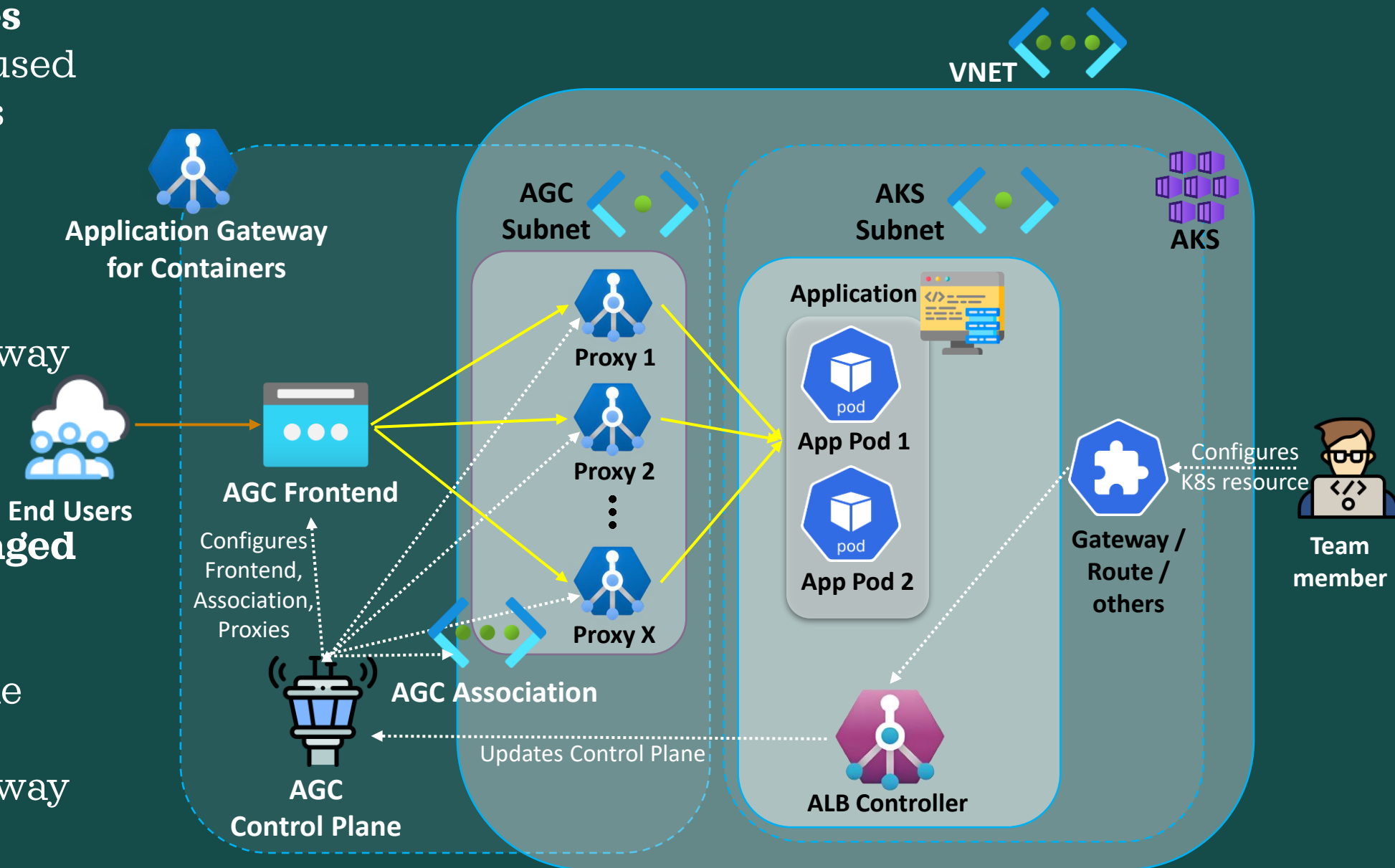
- Abstract; will be used by a set of proxies

Subnet delegation

- Makes the subnet dedicated to Application Gateway for Containers resources

User assigned managed identity

- Allow the ALB controller, to make changes to Application Gateway for Containers



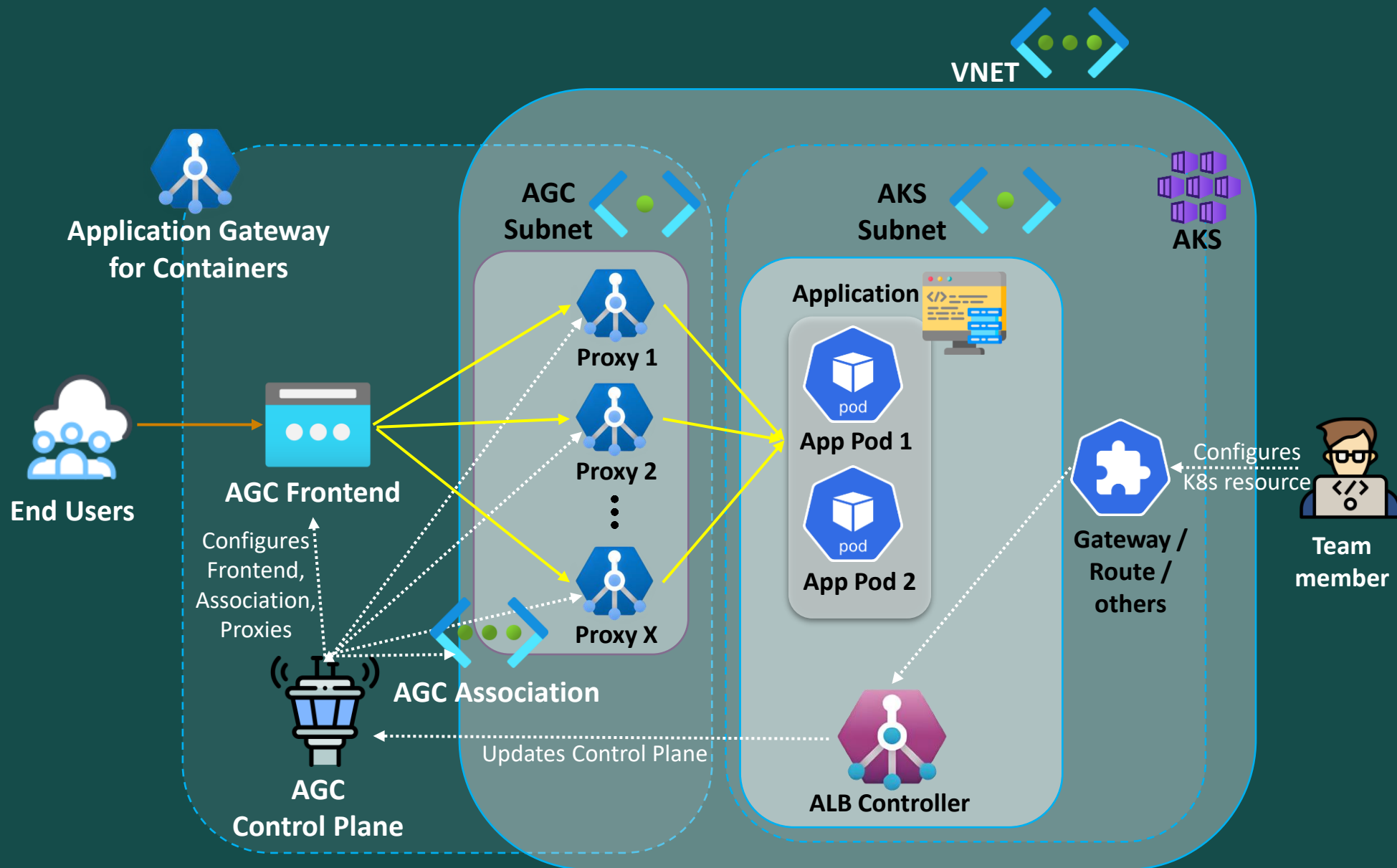


What is Application Gateway for Containers?



Frontends

- Specify where client traffic enters the gateway
- Provide unique FQDN for custom domain, usable with CNAME records
- One Application Gateway for Containers can handle multiple frontends simultaneously



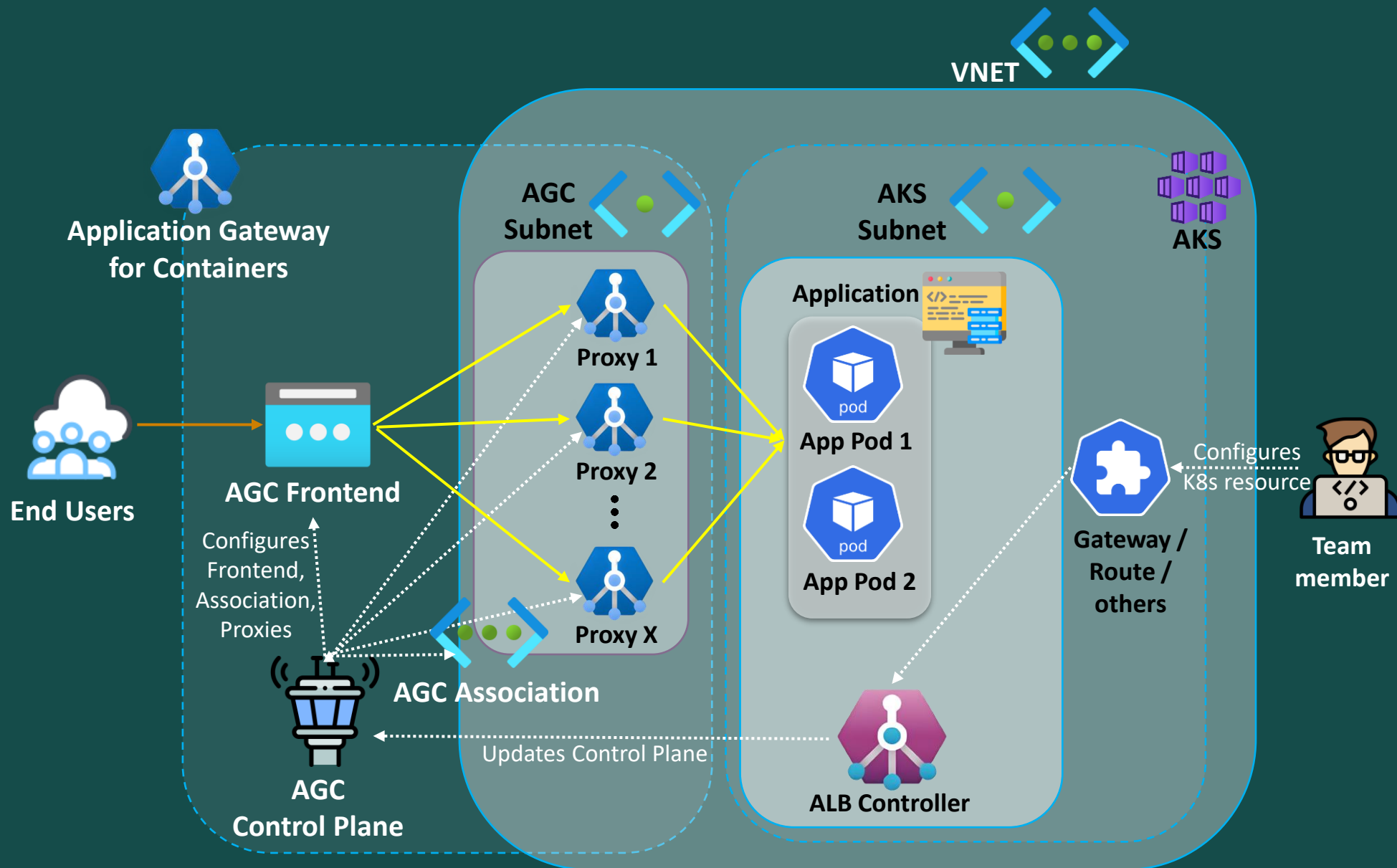


What is Application Gateway for Containers?



Associations

- Establishes a link to a virtual network, serving as a 1:1 mapping between an association resource and a delegated subnet
- A minimum of /24 is needed



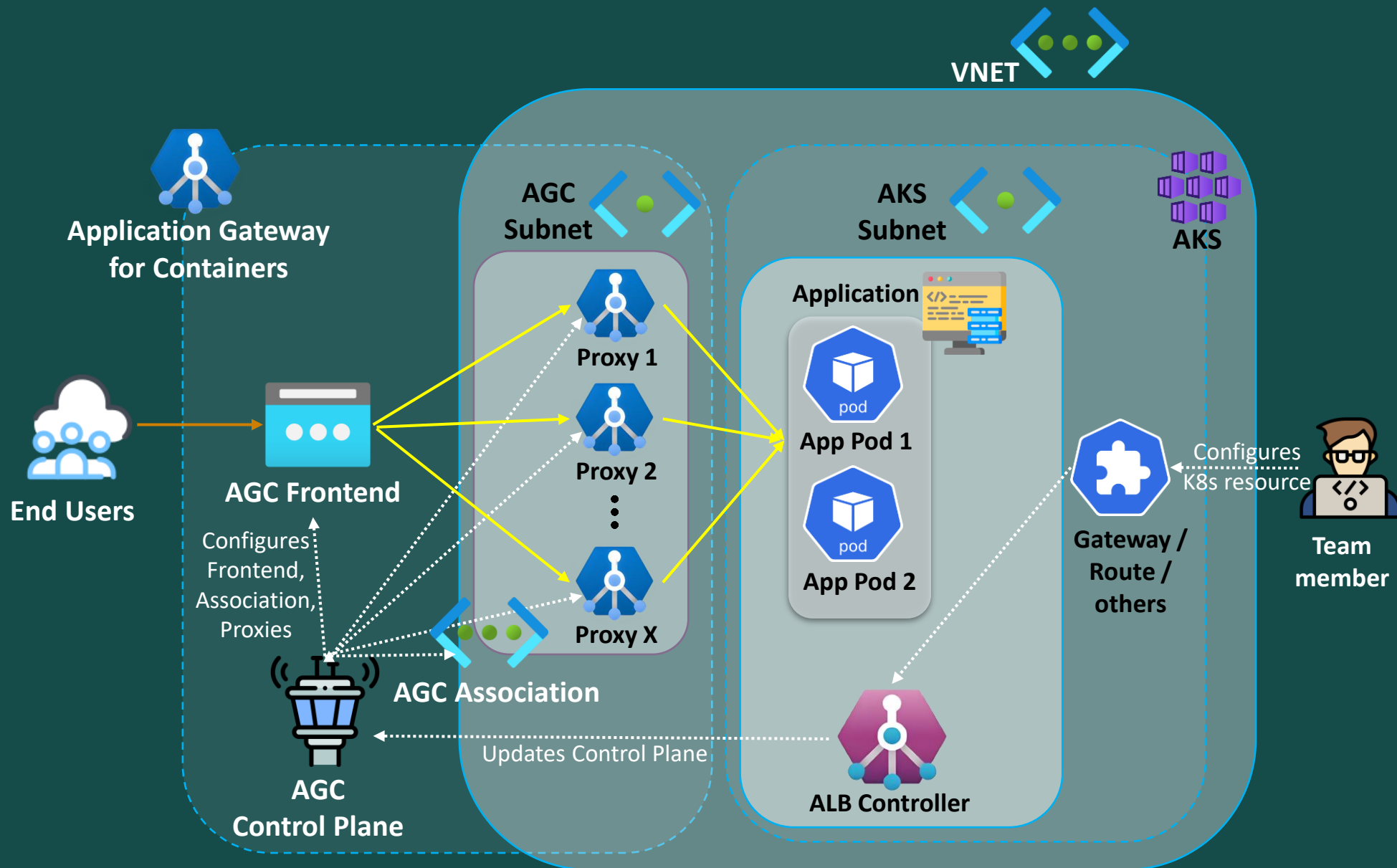


What is Application Gateway for Containers?



Control plane

- Automatically deployed
- Receives updates from the ALB Controller and configures the Frontend, Association and Proxies
- Contains a set of proxies that are injected into the Application Gateway for Containers' subnet



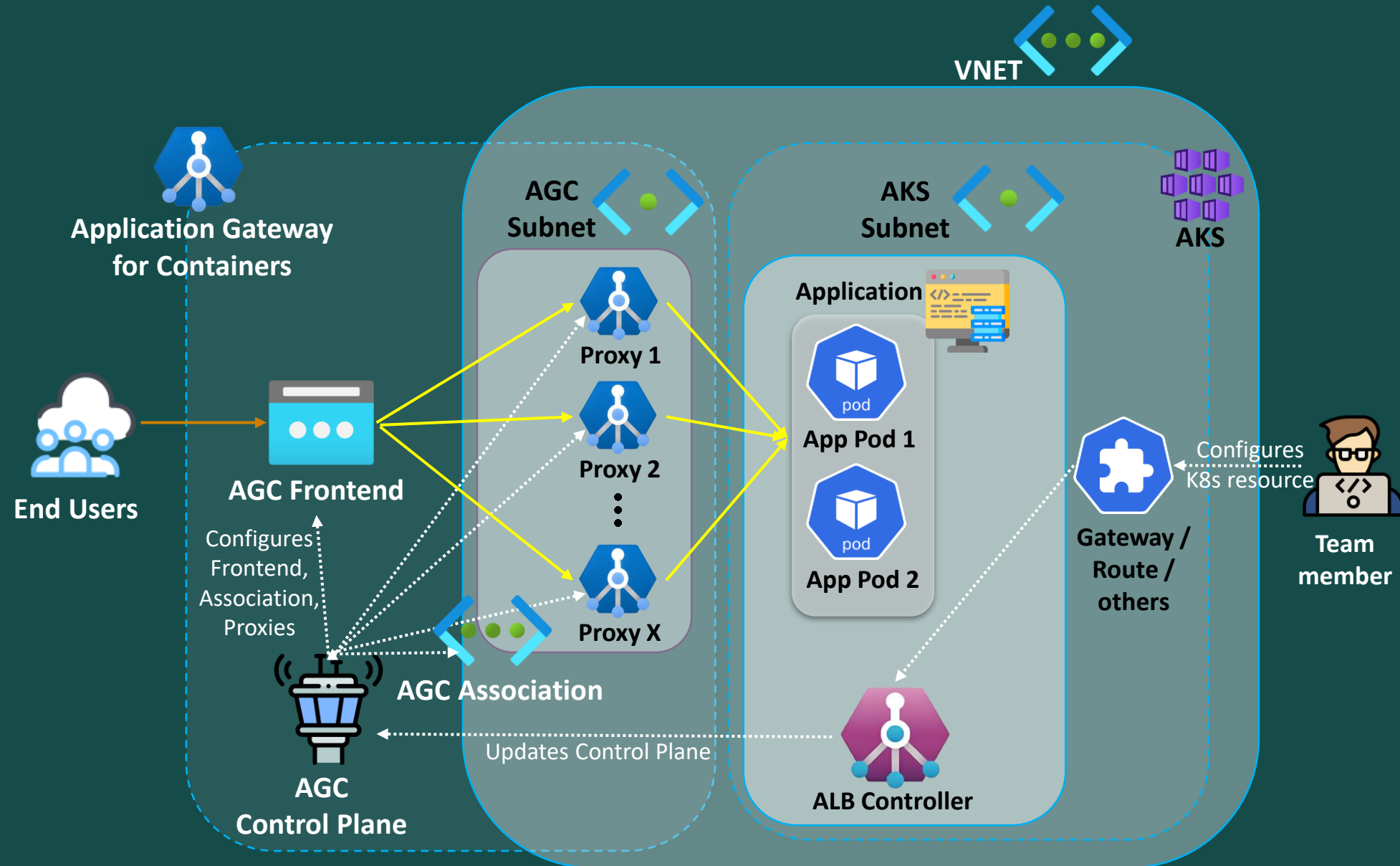


What is Application Gateway for Containers?



Proxy

- Managed by the control plane and are abstracted from the user
- Used to route end-user traffic according to the application pods



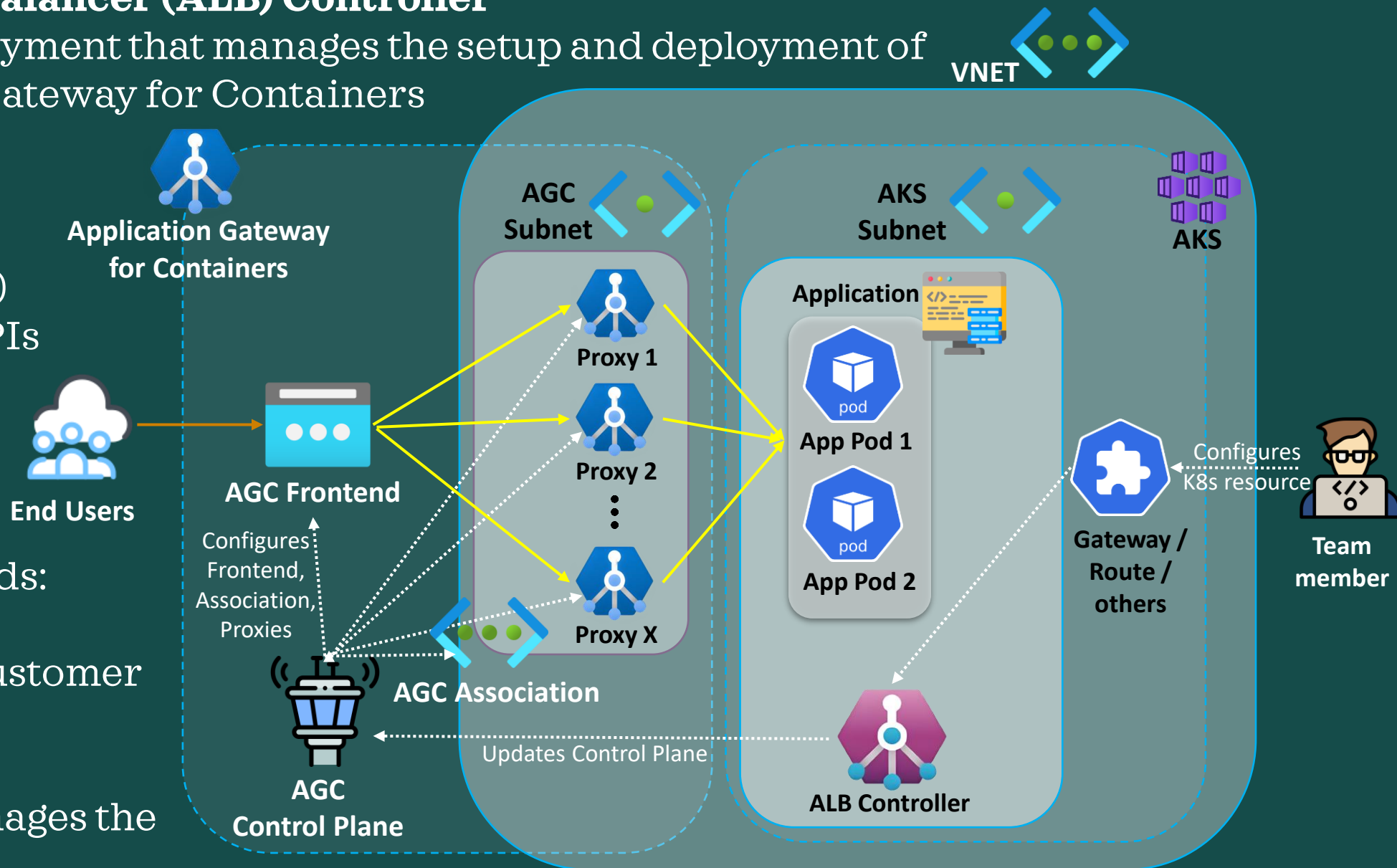


What is Application Gateway for Containers?



Application Load Balancer (ALB) Controller

- Kubernetes deployment that manages the setup and deployment of the Application Gateway for Containers
- Monitors Kubernetes resources (e.g., Ingress, Gateway) and uses ARM/APIs for Application Gateway for Containers configuration
- Consists of two pods:
 - alb-controller: orchestrates customer intent
 - alb-controller-bootstrap: manages the related CRDs





What is Application Gateway for Containers?



- **Deployment strategies**
 - **Bring your own (BYO) deployment:** You can manually create the Application Gateway for Containers, the Frontend(s), and the Association(s), and then reference them in the Kubernetes configuration
 - ARM calls are not used
 - **When using Gateway API**
 - Provision the Frontend first and then reference it in the Kubernetes Gateway resource
 - The Frontend is managed by the Azure admin; won't be deleted when Kubernetes Gateway is removed
 - **Managed by ALB Controller:** ALB Controller manages the lifecycle of the Application Gateway for Containers and sub-resources. The ALB Controller creates the Application Gateway for Containers when the ApplicationLoadBalancer custom resource is defined. “managed-by-alb-controller : true” is set at the Application Gateway for Containers level
 - ARM is used only when there is the need to create
 - **When using Gateway API**
 - ALB Controller automatically creates a Frontend when a Gateway references the ApplicationLoadBalancer. Frontend's lifecycle syncs with Gateway, meaning the deletion of the Gateway removes Frontend